d) Aspartate and glutamate

13. _____ acts as a methyl donor for the synthesis of epinephrine, creatine and thymine.

a) Methotrexate

b) **S- adenosyl methionine**

c) Tetrahydrofolate

d) Biotin

14. Catabolism of phenylalanine and tyrosine yield _____.

a) Succinate

b) α-ketoglutarate

c) Malate

d) **Fumarate**

15. Co-enzyme required for the conversion of L-Amino acid into L-Imino acid.

a) FAD^+

b) **FMN**

c) NAD^+

d) $NADP^+$

II Give short answer for the following

1. What is the fate of excess dietary amino acids?

2. What are non-essential amino acids? How are they synthesized?

3. Give the reaction catalyzed by AST.

4. What are glucogenic amino acids and ketogenic amino acids? Give examples.

5. What is decarboxylation? Give example.

6. Name the enzymes involved in urea cycle.

7. Mention any four amino acids whose carbon skeletons enter into TCA cycle.

8. How is citrulline formed in the mitochondria converted to argininosuccinate in the cytosol?

9. How is tyrosine converted to dopaquinone?

10. What is the role of S-adenosyl methionine in conversion of norepinephrine to epinephrine?

III Give short answer for the following

1. What is transamination? Add a note on its features.

2. Explain the different types of deamination reactions with examples.

3. Write a note on the formation of amines from amino acids.

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